The Ultrametric Program: One Structural Object Across Seven Research Domains, and Its Falsifiable Tests

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Abstract

Seven research domains — ultrametric physics, the laws of form, infomatics, paradigm engineering, consilience research, a cloud-native platform, and interactive demos — are claimed to be seven vocabularies for one structural object: a nested, hierarchical partition logic. The object is defined by the ultrametric inequality and its strict hierarchy of nested balls; the specific arithmetic — p-adic valuation, the adelic product formula — is one realization, the hierarchy is the invariant. This paper states that thesis plainly and makes it testable. The program's scientific content is carried by three falsifiable hypotheses: H1, that ultrametric structure is an effective compression and clustering prior for high-dimensional sparse measurement data; H2, that continuous Archimedean physics appears as the thermodynamic or ergodic average over the leaves of the ultrametric hierarchy; and H3, that quantum-coherent systems under structured hierarchical noise exhibit decoherence scaling that deviates from the standard Markovian prediction in a p-adic pattern. The strong form of the program is bound to a 2028 decision point: if neither H1 nor H3 yields a positive result, the program's claim of a physics-relevant non-Archimedean substrate is falsified. The program serves a mission — an energy-efficiency benchmark for quantum computing (joules per correct solution) — to which the thermodynamic bounds of computation are the direct link. Evidence from the program's own corpus is presented where it exists, including a computational audit of the keyword taxonomy that shows the consilience is semantic rather than lexical, and a deterministic verification suite whose numbers are reproduced exactly by the deposited scripts. The paper also confronts the program's deepest open questions: the observer's resolution hierarchy, the global topology of the tree, and whether the unity is one radix or a family of incommensurable grammars.

1. Introduction: one program or seven?

A research organization that spans number-theoretic physics, a calculus of logical form, the thermodynamics of computation, technology forecasting, measurement epistemology, a software platform, and interactive visualization invites a direct question: is this one program or seven programs that share an organization? This paper answers: the domains are claimed to share one structural object — nested hierarchical partition logic — and the purpose of this paper is to state that claim at its full width, to make it falsifiable, and to connect it to the program's mission.

The claim has three layers, and keeping them separate is the discipline of the paper. First, the structural identification: the domains share a family of nested-partition structures. Second, the scientific content: three testable hypotheses that would give the identification empirical teeth. Third, the mission: an energy-efficiency benchmark for quantum computing, grounded in the same structure. The layers have different epistemic status. The identification is a modeling choice. The hypotheses are empirical claims with disconfirmation criteria and a deadline. The mission is an engineering program that stands or falls on its own metrics.

Why should a reader care? Three reasons. First, the claim is not vacuous: the paper states exactly what would falsify each hypothesis, and it binds the strong form of the program to a 2028 decision point. Second, the program's mission addresses a real and growing need — the energy cost of correct quantum answers — and the paper connects the abstract structure to that mission through the thermodynamic bounds of computation. Third, the paper's evidence discipline is transferable: its computational audits are deterministic, seeded, and deposited with the paper, so every number can be reproduced byte-identically by any reader.

The paper is organized as follows. Section 2 states the program and its mission. Section 3 defines the structural object and its invariant. Section 4 develops the resolution hierarchy of observation — the program's answer to where the observer sits. Section 5 presents the evidence from the corpus, including the keyword-taxonomy audit that established the consilience is semantic, not lexical. Section 6 states the three hypotheses with their disconfirmation criteria. Section 7 gives practitioner deliverables. Section 8 discloses where the premises end. Section 9 positions the claim against the external literature. Section 10 states limitations and open problems. Section 11 discusses the broader significance, and Section 12 documents reproducibility.

2. Program and mission

The QNFO program pursues a specific mission: **an energy-efficiency benchmark for quantum computing** — the question "what does a correct quantum answer cost in energy?" The benchmark (joules per correct solution, JPCUB) is intended as an open, reproducible, energy-first standard across quantum computing platforms: a protocol for measuring the end-to-end energy cost of producing a correct, useful quantum answer, grounded in the physics of computation.

The connection to this paper's structural claim is not decorative. The thermodynamic bounds of computation — Landauer's bound on erasure, the Bremermann limit on computation rate, the Margolus-Levitin bound on evolution speed — are the program's bound family, and they sit naturally inside the hierarchy invariant: each bound is a constraint at a scale, and the bounds nest like the balls of an ultrametric space. The program's mission literature is already substantial: the joules-per-solution metric definition and measurement protocol (10.5281/zenodo.21637028), a system-level comparison of seventeen quantum platforms (JPCUB competitive landscape, corpus record jpcub-competitive-landscape), and a qudit-architecture advantage analysis. Externally, the mission has a peer: Alves, Pezzutto, and Omar (arXiv:2601.03141v2, 2026) benchmark the energetics of Rydberg-atom quantum computing and demonstrate a regime of quantum energy advantage for the Fourier transform against classical supercomputers — the first platform-level energy benchmark of its kind. The joules-per-solution standard is the natural common measure for such results.

The mission is the answer to “what is the program for”. The structural claim is the program’s answer to “what is the program about”. Neither reduces to the other, and this paper does not attempt to make them reduce: the mission stands on its own metrics, the structural claim on its own disconfirmation criteria.

3. The structural object

The invariant is stated in one sentence: **measurement hierarchies organize as nested partitions — a strict hierarchy of nested balls satisfying the ultrametric inequality $d(x,z) \leq \max(d(x,y), d(y,z))$.** The specific arithmetic is one realization: p-adic valuation assigns to a difference its power-of-prime depth, and the adelic product formula assembles all completions into one global object. The hierarchy is what survives a change of base: replace $\mathbb{Q}_p$ with formal Laurent series, tropical semirings, or plain nested partitions without numbers, and the ultrametric inequality and its nested balls remain. The prime-specific arithmetic is accidental; the hierarchy is essential.

The program’s mathematical spine for this identification is the bridge theorem (10.5281/zenodo.21102770), a rigorous framework connecting p-adic and Bruhat-Tits geometries; the Bruhat-Tits tree as the unifying geometric object (R2-distributed record ballistic-transport-on-the-bruhat-tits-tree); and the consilience between physics and number theory (10.5281/zenodo.21590155). The consilience framework record (10.5281/zenodo.21804073) carries the same bridge from valuation theory to the void — the semantic reading of the invariant that the taxonomy audit later confirms. Prior corpus work established that positional notation itself is natively an ultrametric tree (10.5281/zenodo.21046213) and that prime valuation depth reads multiplication as branching (10.5281/zenodo.21918838) — two concrete instances where ordinary mathematics already wears the structure.

The claim at this layer is deliberately modest: the paper does not assert that reality is ultrametric. It asserts that a family of nested-partition structures is the program’s organizing invariant, and that the program’s scientific standing is decided by the hypotheses of Section 6.

4. The resolution hierarchy of observation

Where does the observer sit? The program’s answer — developed in the corpus — is that the observer is a node inside the tree, and that measurement is the resolution operation that moves through the hierarchy. This is the program’s broadest theme, and it deserves the paper’s lead framing: the program is the scientific study of the **resolution hierarchy of observation**.

The corpus anchors are two records. The 29-schisms synthesis (10.5281/zenodo.21458373) formalizes physics as a self-referential calibration problem: the laws are the stable fixed points of mutual consistency between observer, apparatus, and world (the Bootstrap Theorem), and the machinery is ultrametric geometry, Bruhat-Tits trees, the Monna projection, and the syntactic token calculus. The observer-inside-the-tree record (10.5281/zenodo.21473899) scrutinizes the claim that embedding the observer as a node in an ultrametric tree eliminates the need for an external vantage point. Its verdict is the program's theme stated with its boundaries: the resolution survives scrutiny in a limited but genuine sense — the calibration map provides a well-defined internal perspective, and the ultrametric structure avoids the circularity of flat-space relational approaches — but three constraints are non-trivial. First, the distance function requires a global tree topology that is not locally computable, leaving a residual external perspective that the framework acknowledges but does not eliminate. Second, the boson-fermion observer paradox constrains which nodes can simultaneously function as observers. Third, the resolution is observationally indistinguishable from relational quantum mechanics and QBism in currently feasible regimes — it is a structural resolution, not an empirical one.

These constraints are not footnotes; they are the paper's sharpest open problems. The global-topology constraint, in particular, relocates the external perspective rather than eliminating it: the tree itself is an external structure not derivable from any single node's perspective. The program's claim to have resolved the inside/outside schism must therefore be stated as: the schism is relocated to the global topology, where it becomes a mathematical question about the structure rather than a metaphysical one about the observer. That is progress, and it is stated plainly.

5. Evidence from the corpus

The program's claims are not asserted in a vacuum; the corpus of published records provides the evidence base, and this section reports what is established, computationally, at the time of writing.

5.1 The keyword-taxonomy audit

The most direct audit of the program's unity is the computational examination of its own keyword taxonomy (ref 1: docs/QNFO-KEYWORD-TAXONOMY.md v1.0, 2026-08-05 — 335 keywords, seven program sections, three cross-cutting themes), published as the predecessor record (10.5281/zenodo.22071421). The result is a negative result with a positive reading: the taxonomy is strictly partitional — 334 of 335 keywords occur in exactly one program, one keyword occurs in two, none occurs in three or more — and the load-bearing core defined by shared vocabulary is empty (Fisher exact test for bridge-vocabulary enrichment: $p = 1.0$). The consilience, if it exists, is not lexical.

The audit then finds where the consilience does live, at four levels. Semantic families: of the four bridge families, only the hierarchy family spans three programs (laws of form, consilience research, demos); the valuation family sits in ultrametric physics, the distinction family in the laws of form, the bound family in infomatics. The taxonomy's own bridge subsections are program-local anchors that name connections without instantiating shared vocabulary. The taxonomy's explicit cross-cutting themes are where program vocabulary actually meets (the platform and the consilience program). And the published corpus carries the semantic bridges: measurement stratigraphy linking epistemology to valuation theory (10.5281/zenodo.21705220), the valuation-without-reals framework (10.5281/zenodo.21803677), and a computational study finding ultrametric topology in semantic memory with invariant cross-ratio stability (10.5281/zenodo.19564091).

The reading for this paper: the program's unity is a semantic claim about concepts and corpus structure, not a lexical fact about keywords. The taxonomy audit is evidence for that reading, not the headline.

5.2 Computational verification of the hypotheses

The three hypotheses of Section 6 are checked in code before they are asserted, by a deterministic, seeded verification suite (scripts and full logs in the predecessor's deposit; all numbers reproduced byte-identically). H1, retrieval: the data-derived ultrametric index (single-linkage recoding over cosine distances) matches a cosine baseline exactly on a seeded synthetic corpus (precision at 10: 1.000 vs 1.000) and trails by 0.042 at precision at 10 on a 69-title labeled corpus (0.765 vs 0.807); the naive sha256 p-adic-hash variant collapses toward random retrieval (0.210), confirming that the hash encoding is a convention, not physics. H2, numeric: the b-adic tree metric is exactly ultrametric (zero violations over 30,000 triples), and the ergodic mean over leaves converges to the central-limit golden value (relative error 0.004-0.039 against σ^2/n) with Gaussianity confirmed. H3, numeric: the p-adic valuation-suppressed noise model yields decoherence scaling slope -0.9881 against the Markovian -2.0000 — a separation of 1.012 in log-log slope, with the summation arithmetic verified exactly (0.0 relative error) and a seeded Monte Carlo sanity check passing.

The summary, stated plainly: H1 is partial on the two pinned corpora (the abstract-and-embedding corpus specified by the protocol is the adjudicator); H2's numeric machinery is confirmed but no derivation of Archimedean physics from the hierarchy exists yet; H3's signature is real in the model and detectable in principle, and the nearest existing external experiment is a cavity-QED driven-dissipative spin glass showing incipient ultrametric order (arXiv:2307.10176v2, 2023).

6. The three hypotheses and their disconfirmation criteria

H1 — compression prior. Ultrametric structure is an effective compression and clustering prior for high-dimensional sparse measurement data: on at least two independent corpora, an ultrametric index matches or beats a cosine baseline on retrieval precision. *Disconfirmation criterion:* H1 fails if ultrametric retrieval does not match the cosine baseline on two pre-specified corpora with metrics, primes, and hashes committed before measurement. Current state: exact match on the synthetic corpus, -0.042 at precision at 10 on the title corpus; adjudication pending on the abstract-and-embedding corpus.

H2 — Archimedean emergence. Continuous Archimedean physics appears as the thermodynamic or ergodic average over the leaves of an underlying ultrametric hierarchy, in the same sense that smooth hydrodynamics is the average of discrete molecular dynamics. *Disconfirmation criterion:* H2 fails if no derivation exhibits the averaging operation — ergodic mean over leaves or renormalization limit — producing an Archimedean limit theory. Current state: the corpus contains the closest sibling — finite-distinction quantum mechanics (10.5281/zenodo.22046458), which derives unitary evolution and superposition as the large-distinction limit of stochastic thermodynamics — and the external literature supplies the machinery: Markov processes on ultrametric spaces embeddable into $\mathbb{Q}_p$ reduce to Kolmogorov-Feller pseudo-differential equations on $\mathbb{Q}_p$ (Bikulov and Zubarev, arXiv:1504.03629v1, 2015), with m-adic fractional-time random walks as diffusive limits (Dolgopolov and Zubarev, arXiv:1012.1248v2, 2010) and p-adic Gibbs measures and phase transitions on trees (Mukhamedov, Rozikov, and Mendes, arXiv:math-ph/0512018v2, 2005). The derivation target is named; the derivation does not yet exist.

H3 — non-Archimedean signature. Quantum-coherent systems under structured hierarchical noise exhibit decoherence scaling that deviates from the standard Markovian prediction in a p-adic pattern (power-of-prime hierarchy). *Disconfirmation criterion:* H3 fails if structured-noise decoherence measurements show no deviation from Markovian models at the precision of the stated protocol. Current state: the p-adic noise model gives $\tau \sim 1/n$ (slope -1) against the Markovian $\tau \sim 1/n^2$ (slope -2), computationally verified; the corpus contains a platform proposal for p-adic quantum metrology with passive error resilience (10.5281/zenodo.21748299); the nearest external experiment is the cavity-QED spin glass (arXiv:2307.10176v2, 2023).

2028 decision point. The strong form of the program — a physics- relevant non-Archimedean substrate — is falsified by 2028 if neither H1 nor H3 yields a positive result. The decision point is administrable: H1's adjudication corpus is specified by its protocol; H3's measurement protocol is a deliverable (Section 7). A null on both by 2028 ends the strong-form claim; the engineering deliverables and the semantic-consilience reading survive regardless, because they do not depend on the strong form.

7. Practitioner deliverables

Three artifacts make the program usable without any commitment to its ontology, plus one mission deliverable.

Deliverable 1 — the data-derived ultrametric index. A retrieval index built by re-coding a corpus into its single-linkage hierarchy over cosine distances, benchmarked against a cosine baseline on two pinned corpora; the p-adic hash variant is retained as the encoding control. Usable today as a content-addressing and retrieval tool; its performance result is the H1 test.

Deliverable 2 — the structured-noise decoherence protocol. A measurement specification for H3: qubit coupled to hierarchical noise, the noise model, the pulse sequence, the expected scaling (p-adic power-of- prime versus $1/n^2$

Markovian), the significance threshold, and platform notes for trapped-ion and superconducting hardware, citing the cavity-QED experiment as the nearest existing platform. An experimental group can cost this protocol directly from the paper.

Deliverable 3 — the machine-readable consilience map. The graph output of the taxonomy audit (342 nodes, 336 ownership edges) recording programs, keywords, load-bearing flags, and bridge-family memberships — the vocabulary index for the corpus.

Mission deliverable — the JPCUB link. The role of the bound family in the energy benchmark stated in engineering terms: each thermodynamic bound as a scale constraint, the bounds nesting like ultrametric balls, and the joules-per-solution metric as the common measure across platforms — connecting this paper's structure to the program's purpose.

8. Where the premises end

- **L0 — unanalyzable primitives:** the act of distinction (the mark); the notion of observation or measurement; the rational numbers as a field. Nothing below this layer is derived.
- **L1 — imported theorem:** Ostrowski's classification — every nontrivial absolute value on the rationals is Archimedean or p-adic. Used, not re-proven.
- **L2 — structural bridge (named input):** the identification of measurement hierarchies with ultrametric valuation structure. Prior records support this as a correspondence; it is a modeling choice, not a theorem.
- **L3-L5 — hypotheses H1, H2, H3:** empirical claims decided by the criteria of Section 6.

The thesis is as deep as L2; L2 is a premise, not a result. In particular, the paper does not assert that reality is ultrametric. It asserts that a specific compression prior is testable, that a specific emergence claim has a named derivation target, that a specific noise signature is detectable, and that the program's standing is decided by the test.

9. Related work

Ultrametric data science. Murtagh's program is the empirical foundation: ultrametric embedding for data fingerprinting and fast clustering (arXiv:math/0605555v2, 2006); pervasive ultrametricity in high-dimensional and sparse data (arXiv:physics/0702064v1, 2007); ultrametricity measured in text corpora (arXiv:1201.2719v3, 2012); p-adic or ultrametric data modeling (arXiv:0809.0492v1, 2008); and ultrametric logic in data analysis (arXiv:1008.3585v1, 2010). Chehreghani and Chehreghani (arXiv:1812.09225v4, 2018) provide dendrogram-based representation learning, a required H1 baseline; Ganea, Becigneul, and Hofmann (arXiv:1804.01882v3, 2018) provide the hyperbolic-embedding counterpart for hierarchical data.

Ultrametricity in statistical physics. The canonical physics instance is replica symmetry breaking in spin glasses: Parisi's order parameter (Phys. Rev. Lett. 50, 1946, 1983) and the Rammal-Toulouse-Virasoro review (Rev. Mod. Phys. 58, 765, 1986). Recent work has moved from theory to measurement: the overlap distribution in random lasers (arXiv:2209.03781v2, 2022); incipient ultrametric order in a driven- dissipative cavity-QED quantum spin glass (arXiv:2307.10176v2, 2023); and ultrametric Parisi matrices from real-time Keldysh dynamics (arXiv:2406.05842v3, 2024) — a genuine dynamics precedent for H2's averaging requirement. The counterpoint is real: Newman and Stein argue that replica symmetry breaking cannot be correct for finite-dimensional short-range spin glasses (arXiv:cond-mat/0105282v3, 2001). H2 and H3 confront this controversy explicitly rather than ignore it.

Ultrametric stochastic processes. The H2 machinery exists: stationary Markov processes on ultrametric spaces embeddable into $\mathbb{Q}_p$ reduce to Kolmogorov-Feller pseudo-differential equations on $\mathbb{Q}_p$ (Bikulov and Zubarev, arXiv:1504.03629v1, 2015); m-adic stochastic processes and fractional-time random walks have diffusive limits (Dolgopolov and Zubarev, arXiv:1012.1248v2, 2010); p-adic Potts models on Cayley trees exhibit phase transitions (Mukhamedov, Rozikov, and Mendes, arXiv:math-ph/0512018v2, 2005). These are the mathematical precedents for "average over the leaves".

Energy benchmarking. The mission's external peer: Alves, Pezzutto, and Omar (arXiv:2601.03141v2, 2026) benchmark Rydberg-atom quantum computing energetics and find a quantum energy advantage regime for the Fourier transform; Desislavov, Martinez-Plumed, and Hernandez-Orallo (arXiv:2109.05472v2, 2021) document compute and energy trends in deep learning inference — the classical-side context for joules-per-solution.

p-adic and adelic physics. The classical literature (Vladimirov, Volovich, Zelenov, p-Adic Analysis and Mathematical Physics, World Scientific, 1994) is the mathematical foundation; the program's own adelic synthesis record (10.5281/zenodo.21590155) applies it to quantum field theory at the level of toy models.

10. Limitations and open problems

The standing dangerous question. The program's own ignorance audit (Q9 of the Universal Ignorance Audit artifact) poses the threat directly: is the ultrametric program a sophisticated exercise in imposing a beautiful, self-consistent, but ultimately untestable meta-structure, where the rigor of the mathematics masks the absence of a new coupling constant or prediction? The audit does not answer this question; it records it as the standing threat, and the 2028 decision point is the deadline for the positive result that would answer it. This paper adopts the same posture.

The global topology of the tree. The observer-inside-the-tree record's sharpest constraint is also this paper's sharpest open problem: the distance function requires a global tree topology that is not locally computable. The resolution hierarchy of observation therefore carries a residual external perspective — the tree itself. The paper states this plainly: the program's answer to the inside/outside schism is to relocate it to the global structure, where it is a mathematical question, not to eliminate it.

The CFE gap. The paradigm-engineering domain (48 keywords) contains no bridge-family vocabulary and shares no keywords with any other domain in the taxonomy audit. The program's consilience table either builds the CFE bridge explicitly — forecasting and learning-curve keywords as a hierarchy over paradigms — or marks CFE as the weakest documented link. This paper records the gap rather than resolving it.

Encoding dependence. The p-adic valuation of a measurement requires digitizing and hashing the raw reading first; the hash is a chosen convention, not physics. H1's protocol commits the hash, the prime, and the corpora before measurement; the RQ2 result from the predecessor audit is the direct empirical instance — raw-hash p-adic prefixes do not identify consilience links better than cosine at matched pair counts on either corpus, exactly as this limitation predicts.

The dynamics gap. The corpus is rich in statics (geometry, bounds, hierarchies) and poor in dynamics. H2 requires an explicit averaging operation; none is specified in the corpus. The disconfirmation criterion for H2 is written to require that derivation rather than permit it to be assumed.

Plurality. The program's deepest open question is whether its unity is one radix or a family of incommensurable grammars with translation but no reduction. The audit's strongest positive finding — the hierarchy family spanning three domains — supports the invariant-as-hierarchy reading, not a single-radix reading. The vocabulary offers no evidence of one hidden basis; it offers evidence that nested partitions recur. The deliverable-3 map is designed so that either answer can be read off the corpus.

11. Discussion: the broader significance

The program's broadest claim can now be stated with its full width and its boundaries intact. The program is the scientific study of the resolution hierarchy of observation: measurement organizes as nested partitions; the hierarchy is the invariant; the arithmetic is one realization; the observer is a node inside the tree; and the mission is an energy-efficiency benchmark for quantum computing, connected to the structure through the thermodynamic bounds of computation.

What would success look like? For H1: an ultrametric index that matches or beats the cosine baseline on the adjudication corpus, making the compression prior a delivered engineering fact. For H2: a derivation, using the Kolmogorov-Feller machinery, of an Archimedean limit theory from an ultrametric base — the named target. For H3: a structured-noise decoherence measurement showing the p-adic scaling signature, on the cavity-QED platform or another. For the mission: a joules-per-solution benchmark accepted as the common energy measure across platforms, with the Rydberg energetics result as an early peer.

What would failure look like? The 2028 decision point is the answer: no positive result from H1 or H3 by then falsifies the strong form. The engineering deliverables and the semantic-consilience reading survive that falsification; the program's posture is to say so in advance.

The plural-radix question is the program's own challenge to itself. The taxonomy audit shows the single-radix reading is not lexically visible; the hierarchy family is the only bridge family spanning three domains. The paper's position: the invariant is hierarchical partition logic, and whether the program is one tree or a forest of incommensurable grammars is left as the open question the corpus can answer — with the deliverable-3 map as the instrument. A program that can state its own deepest open question in its own publication is not a program hiding from scrutiny; it is a program that has made its structure inspectable.

12. Reproducibility

All quantitative claims in this paper are produced by the deterministic, seeded verification suite archived in `artifacts/verification/` (inherited from the predecessor record, 10.5281/zenodo.22071421, and re-verified on this branch — every regenerated result is byte-identical to the inherited JSONs; the expected outputs are archived alongside): rq5_keyword_load.py (taxonomy audit), rq1_retrieval_benchmark.py (H1), rq2_consilience_links.py (consilience-link test), rq3_archimedean_limit.py (H2 numeric), rq4_noise_scaling.py (H3 scaling). All scripts are pure Python standard library, fixed seed 20260823, no random seeds required beyond the declared constants; re-running from the repository root regenerates every JSON artifact byte-identically. Corpus statistics (8,325 nodes; 1,661 papers — including the predecessor record, RES.022) were read from the program's knowledge-graph endpoint on 2026-08-23. External-literature evidence files (arXiv) are archived in the predecessor's deposit and this paper's artifacts. Runtime: under two minutes for the full suite on the reference machine; no external services required.

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